World Model

The Intelligent Brain of the Next-Generation Autonomous Systems

Presented By: **Mohamed** Abdelfattah (Os)

Slides by **Wuyang** Li

April 23, 2026

What is World Model?

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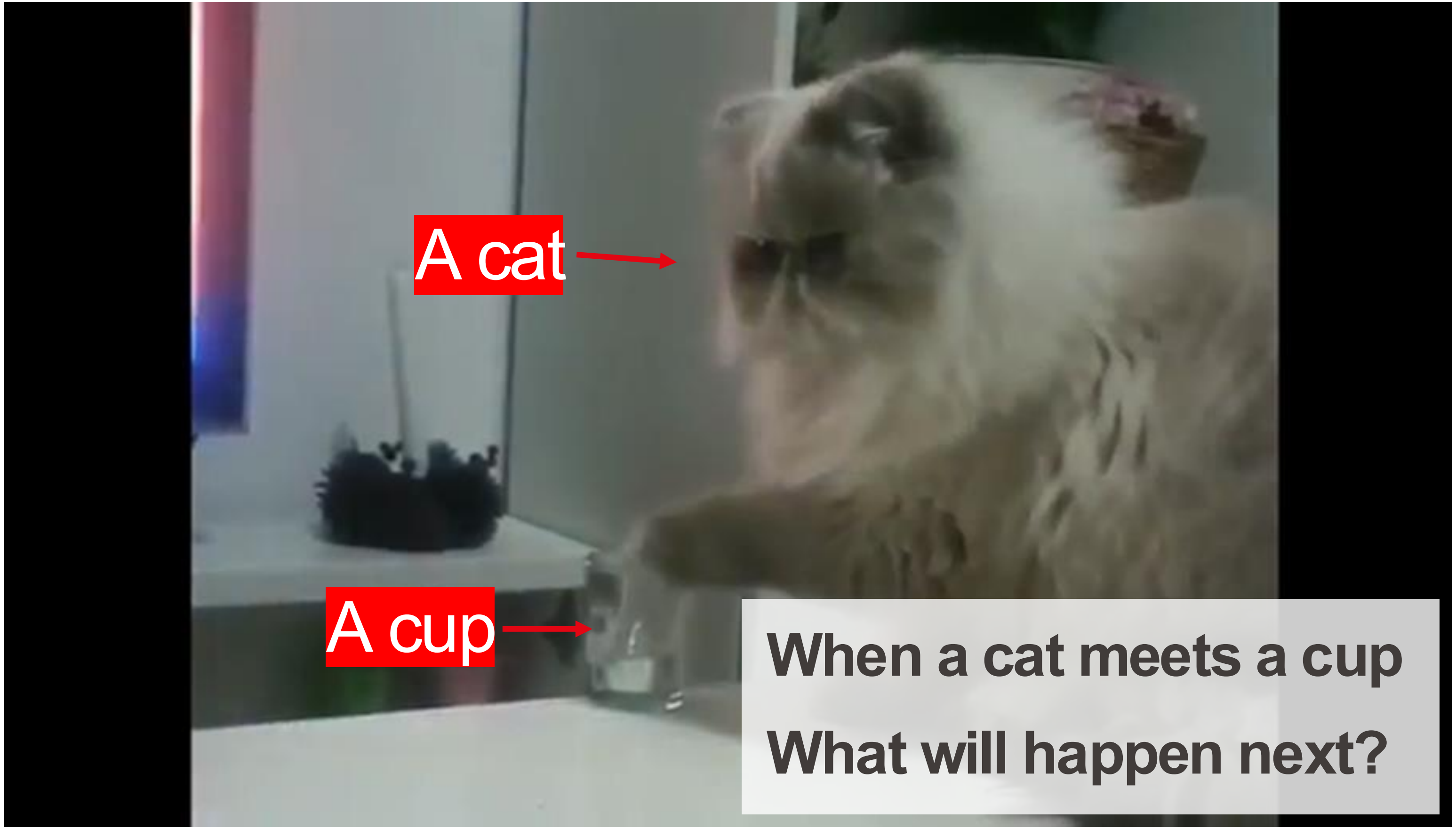

https://www.youtube.com/watch?v=R4anpxoHkPI

What is World Model?

https://www.youtube.com/watch?v=R4anpxoHkPI

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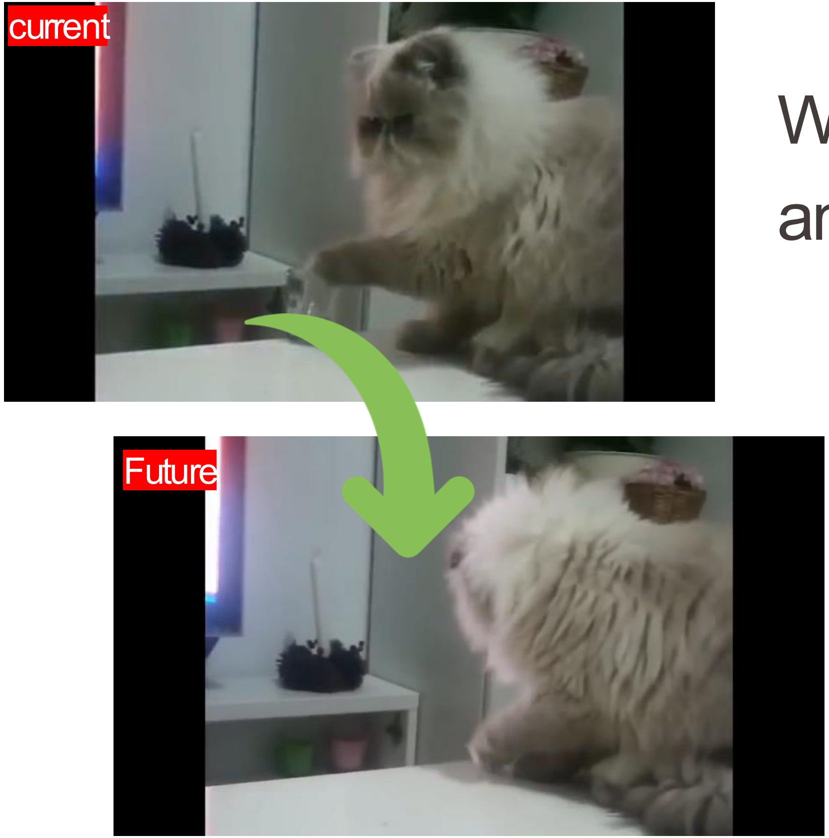
What is World Model?



When you see the current scene and potential action

You can imagine the future

What is World Model?

Why we have such ability？

What is World Model?

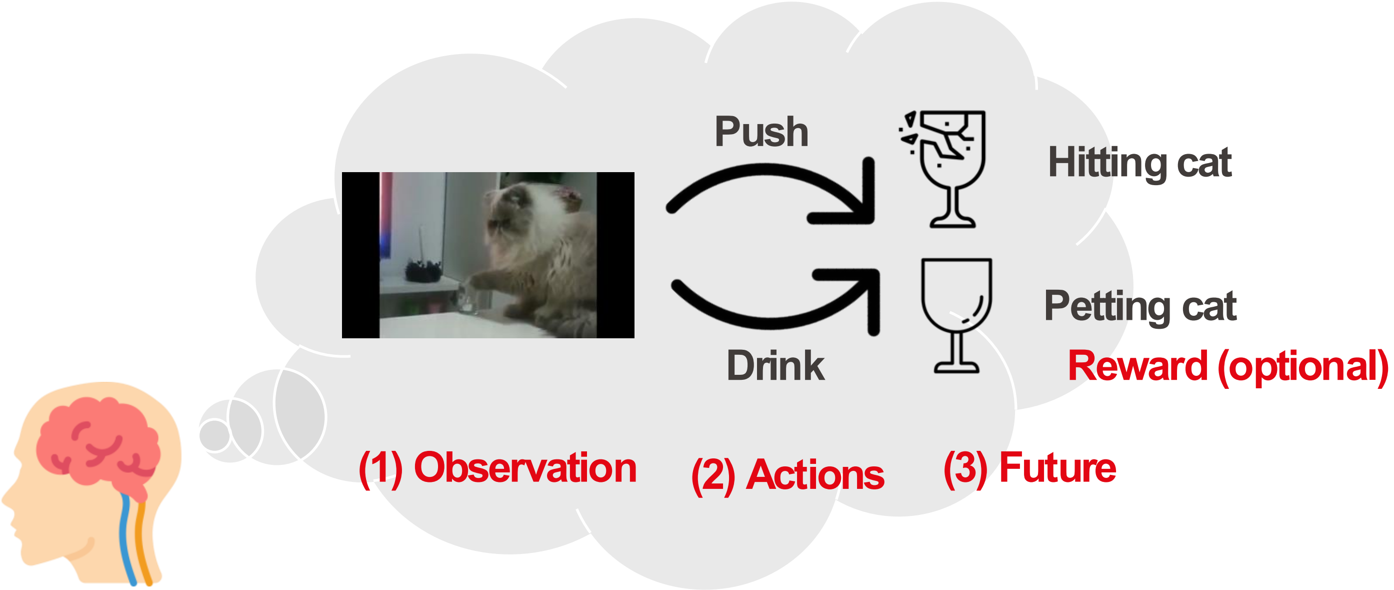
Why we have such ability？

Because our brain secretly has a “mental world”

What is World Model?

Why we have such ability？

Because our brain secretly has a “mental model”



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What is World Model?

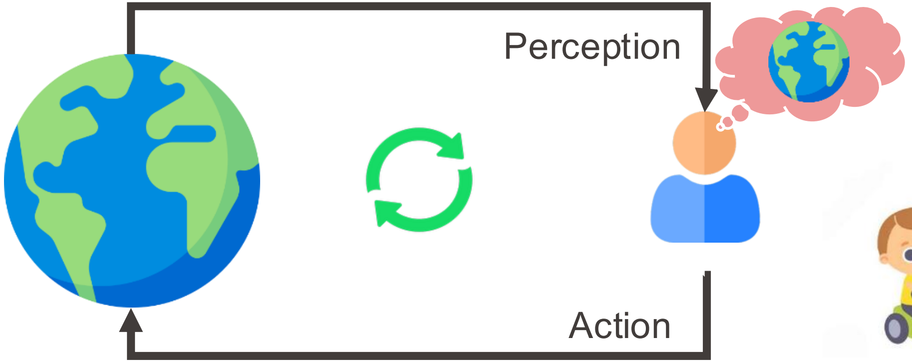
Why we have such ability？

Predict the Future given the Current Observation and Action

1 Delving into World Model

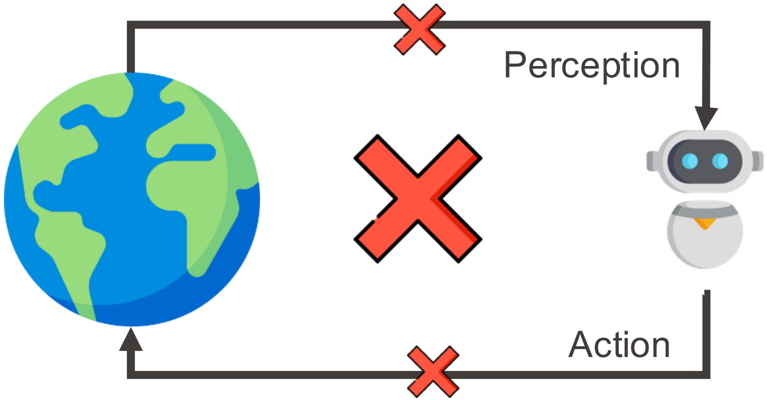
World Model

- Humans learn best by actively engaging with the world.



World Model

- Humans learn best by actively engaging with the world.
  - It is challenging for robots



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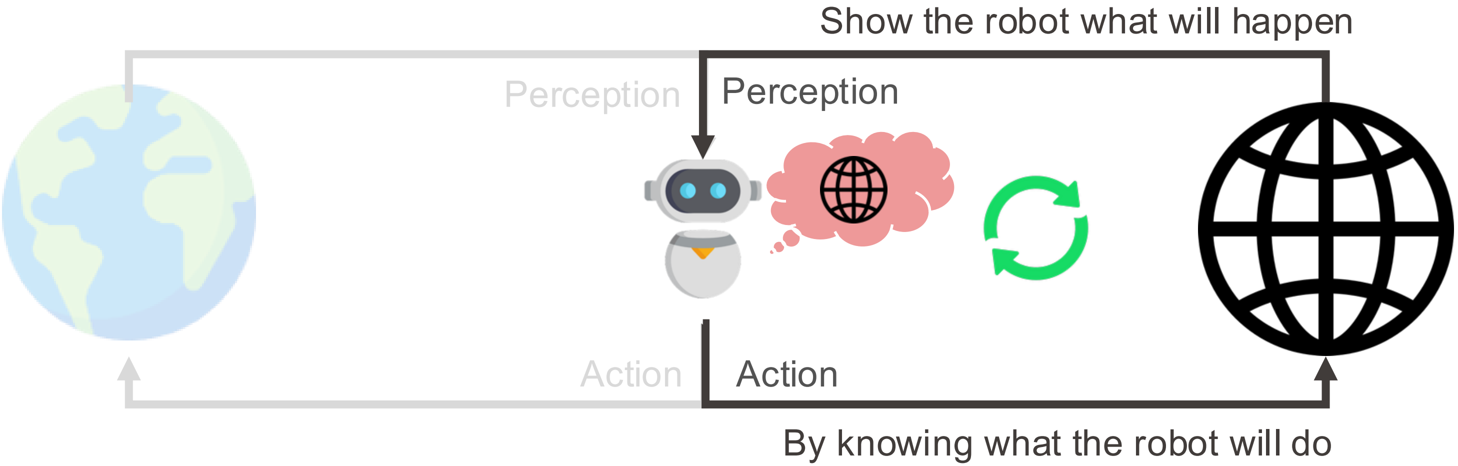
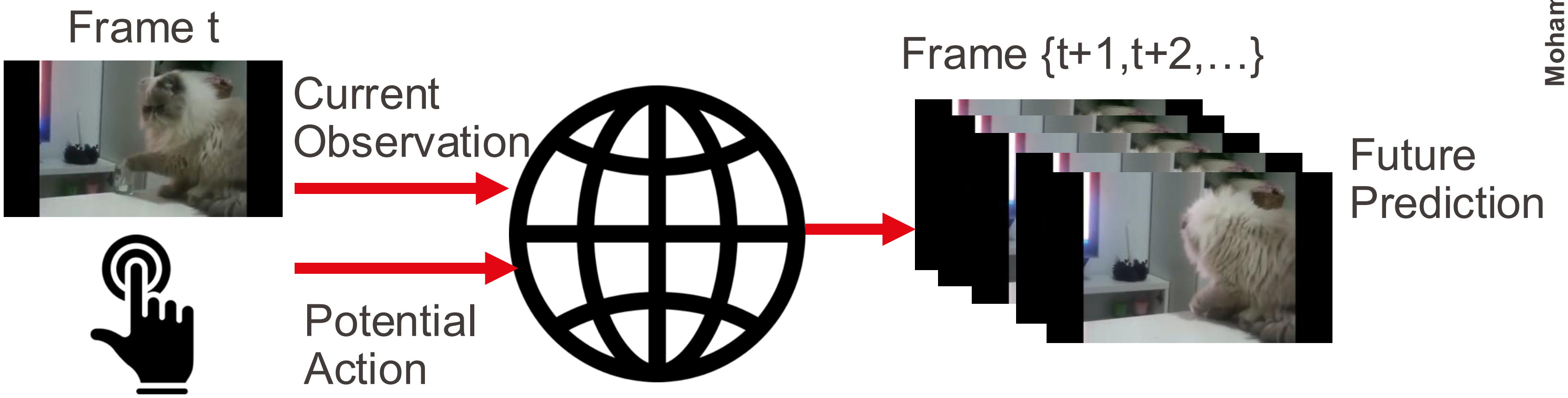
World Model

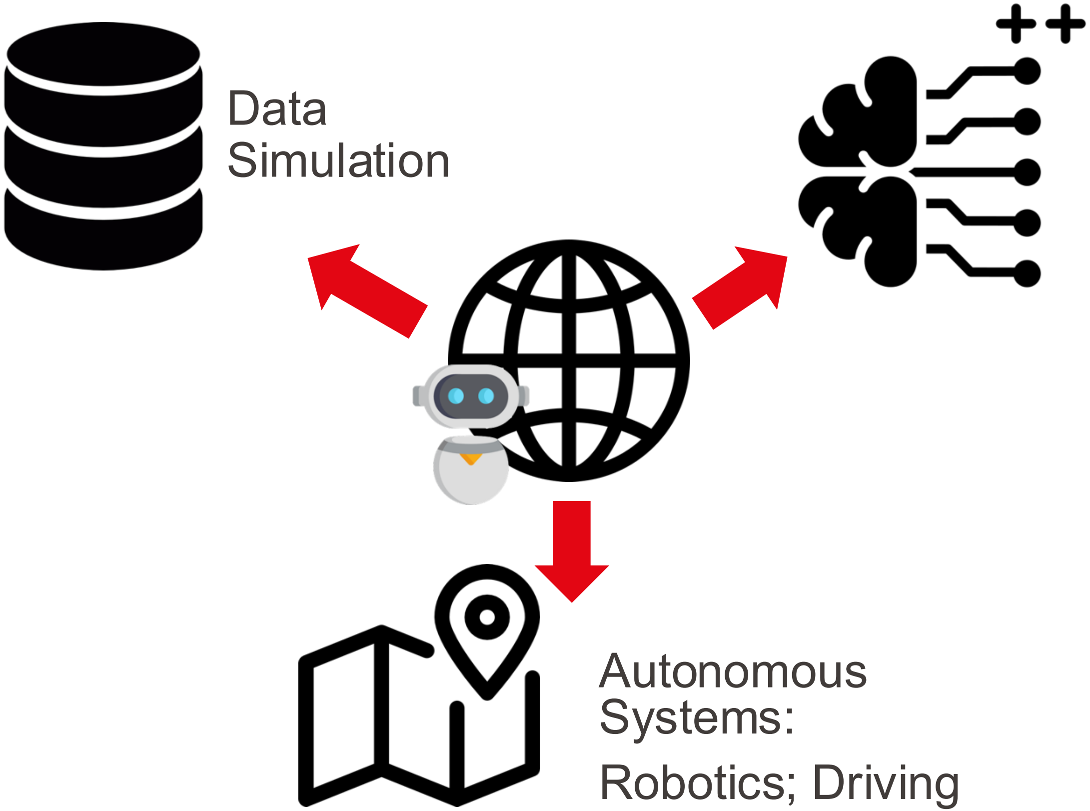
- Humans learn best by actively engaging with the world.
  - It is challenging for robots
- How about simulating a world for robots?



World Model


Frame t
Current
Observation
Potential
Action
Frame {t+1,t+2,…}
Future
Prediction

With World Model, We CAN…



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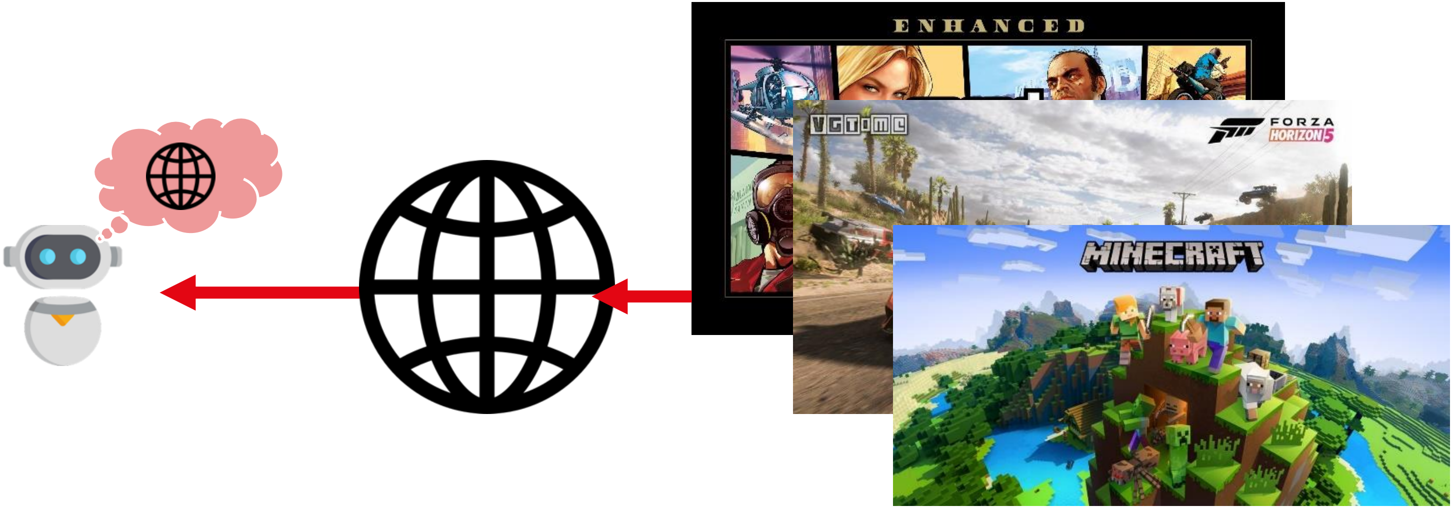
With World Model, We CAN…

1 Delving into World Model

2 Classic World Model

Classic World Model

- Developing an autonomous player to achieve the goal in game



Taxonomy

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- Dreamer [1]
  - Explicit model predicting the future image and reward

- TD-MPC [2] (Temporal Difference Learning for Model Predictive Control)
  - Explicit model predicting the future state and reward

- MuZero [3]
  - Implicit model only making decisions

[1] Hafner et al. Dream to control: Learning behaviors by latent imagination. ICLR 2020

[2] Hansen et al. Temporal difference learning for model predictive control. ICML 2022

[3] Schrittwieser et al. Mastering atari, go, chess and shogi by planning with a learned model. Nature 2020.

Dreamer: Autonomous Player

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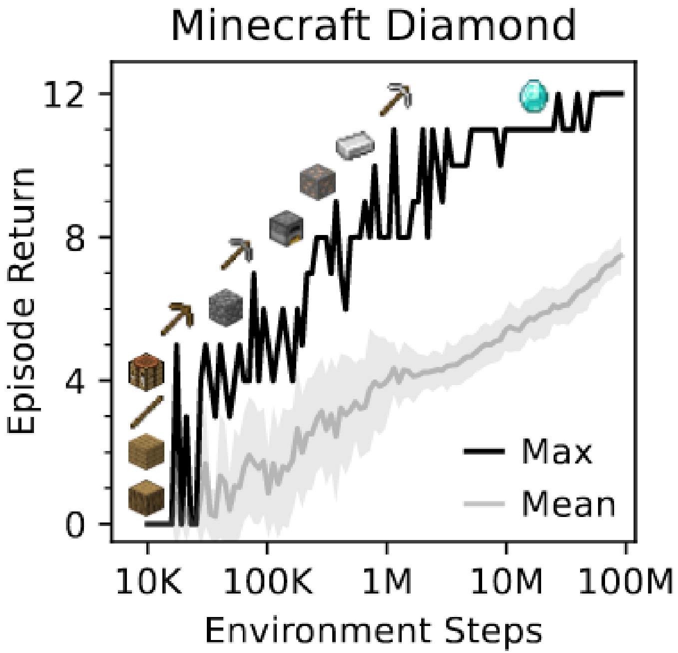
- DreamerV3 is the first algorithm that can collect diamonds in Minecraft without human data (Nature 2025)



Hafner et al. Mastering diverse domains through world models. Nature 2025

Dreamer: Autonomous Player

Step1: World Learning

Train a world model as brain to predict the future

Step2: Policy Learning

Based on the world model, learn policy with reinforcement learning

Dreamer: World Learning

Step1: World Learning

Train a world model as brain to predict the future

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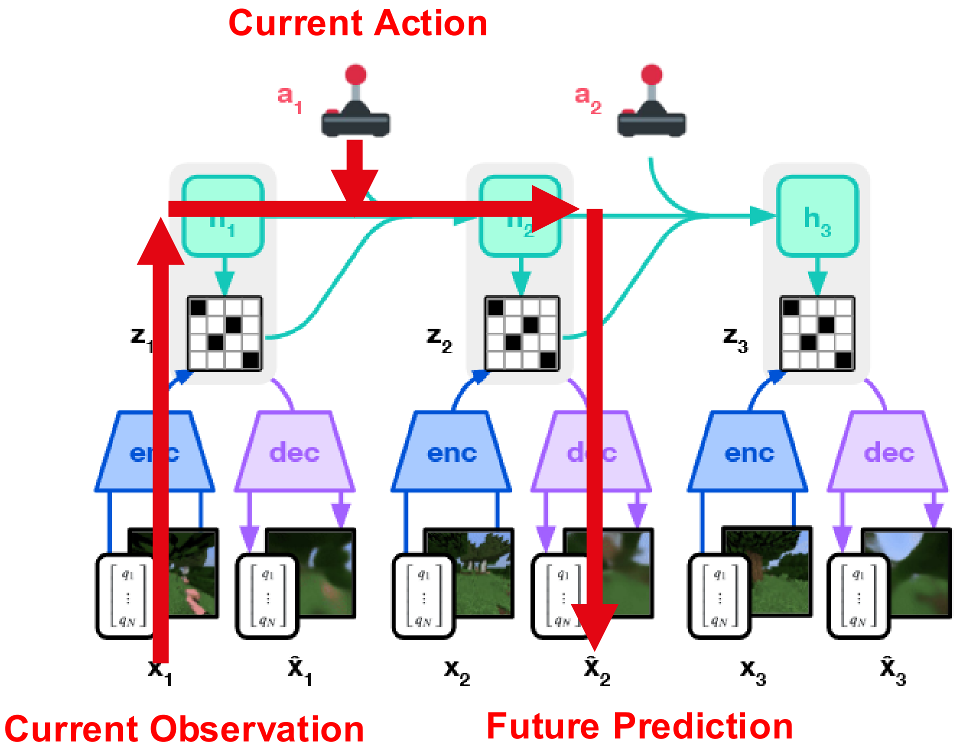
Overview of Dreamer V3



Hafner et al. Mastering diverse domains through world models. Nature 2025

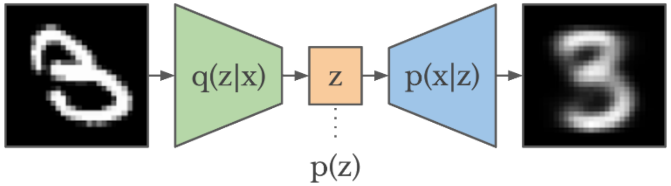
Dreamer: Autoencoder

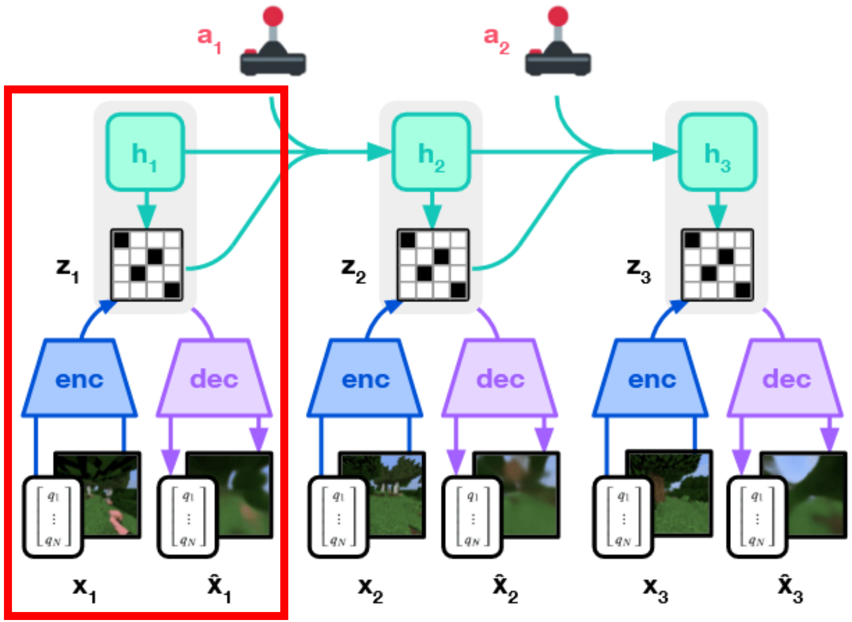
- Encoder & Decoder
  - Encode images into latent

$$z_t \sim q_\phi(z_t \mid h_t, x_t)$$

  - Decode images from latent

$$\hat{x}_t \sim p_\phi(\hat{x}_t \mid h_t, z_t)$$





Hafner et al. Mastering diverse domains through world models. Nature 2025

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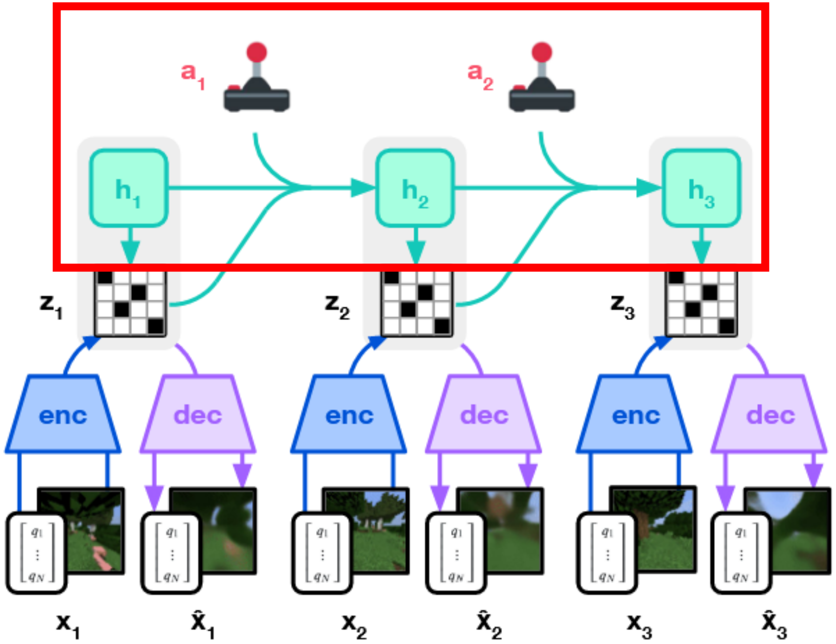
Dreamer: Sequential Model

- Encoder & Decoder
  - Encode images into latent

    $$z_t \sim q_\phi(z_t \mid h_t, x_t)$$
  - Decode images from latent

    $$\hat{x}_t \sim p_\phi(\hat{x}_t \mid h_t, z_t)$$
- Sequential Model
  - Predict the future state

    $$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$$



Hafner et al. Mastering diverse domains through world models. Nature 2025

EPFL

Dreamer: Predictor

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- Encoder & Decoder
  - Encode images into latent

$$z_t \sim q_\phi(z_t \mid h_t, x_t)$$

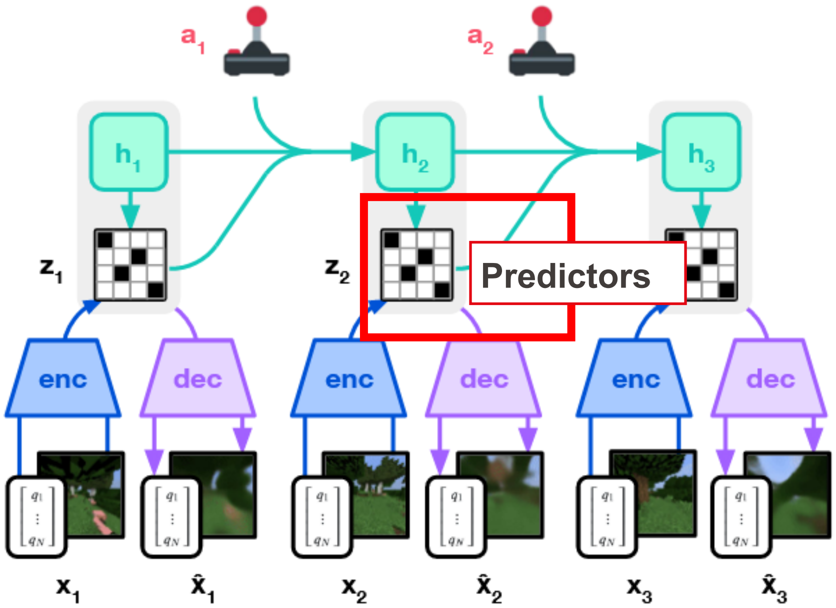
  - Decode images from latent

$$\hat{x}_t \sim p_\phi(\hat{x}_t \mid h_t, z_t)$$

- Sequential Model
  - Predict the future state

$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$$

- Predictors
  - Reward predictor $\hat{z}_t \sim p_\phi(\hat{z}_t \mid h_t)$
  - Continue predictor $\hat{c}_t \sim p_\phi(\hat{c}_t \mid h_t, z_t)$



Hafner et al. Mastering diverse domains through world models. Nature 2025

Dreamer: Policy Learning

Step1: World Learning

Train a world model as brain to predict the future

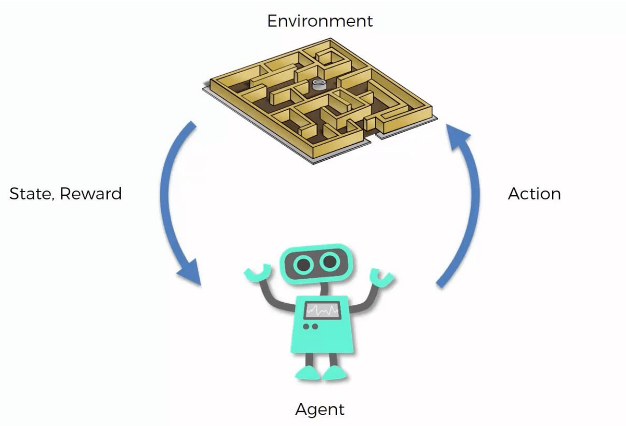
Step2: Policy Learning

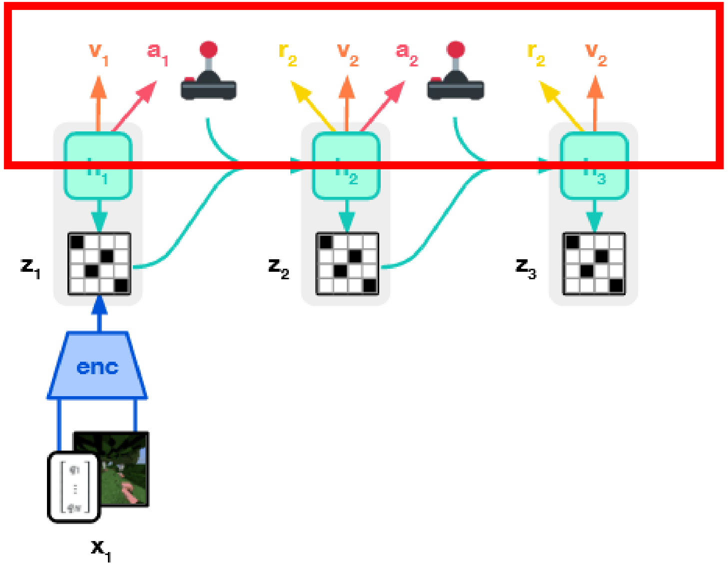
Based on the world model, learn policy with reinforcement learning

Hafner et al. Mastering diverse domains through world models. Nature 2025

Dreamer: How to Use World Model

- Based on the trained world model
  - Predict the future action
  - Predict the value of future state
- Optimize via reinforcement learning



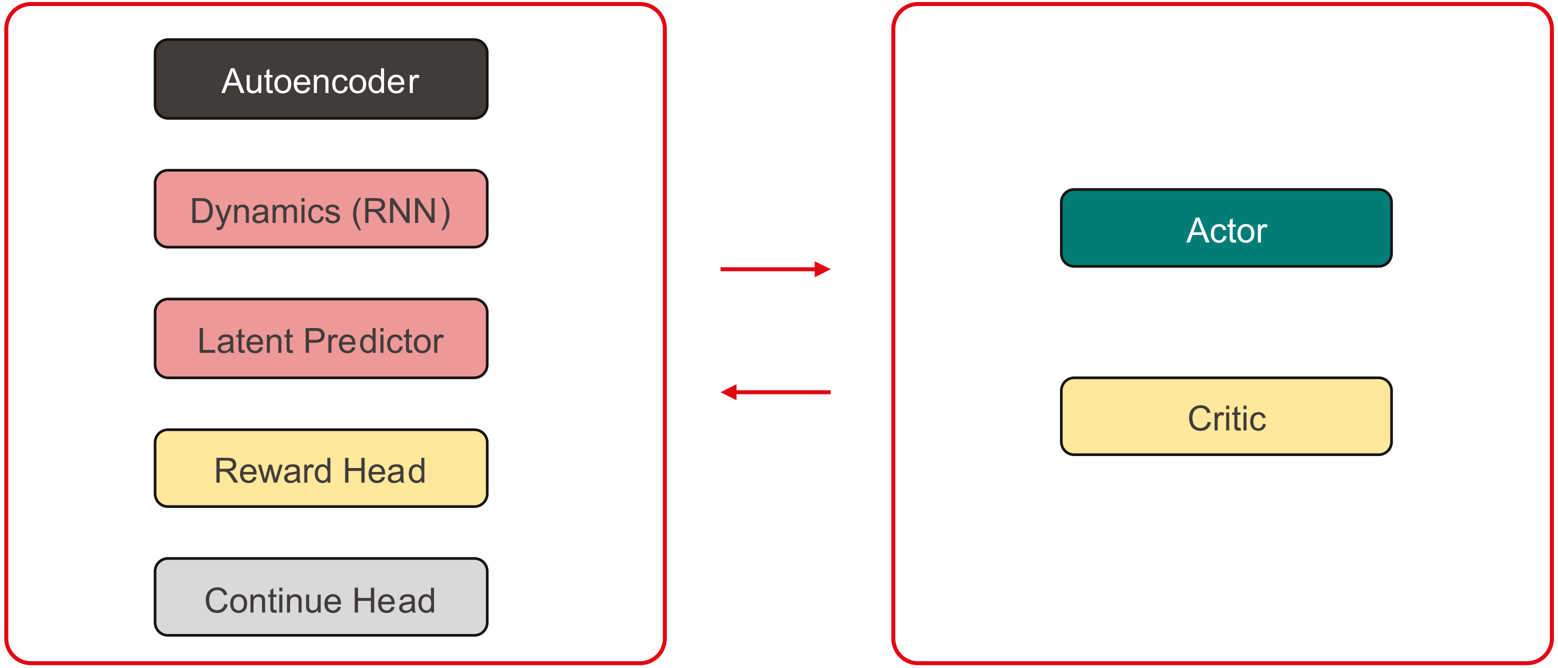


Hafner et al. Mastering diverse domains through world models. Nature 2024

Summary

World Model

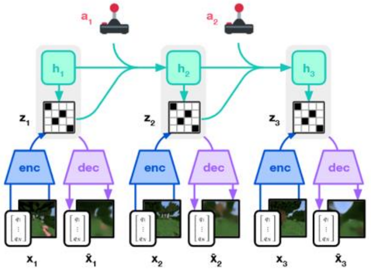
Policy

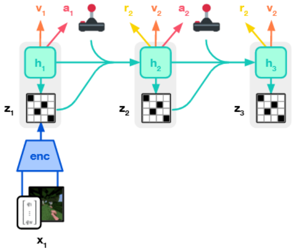


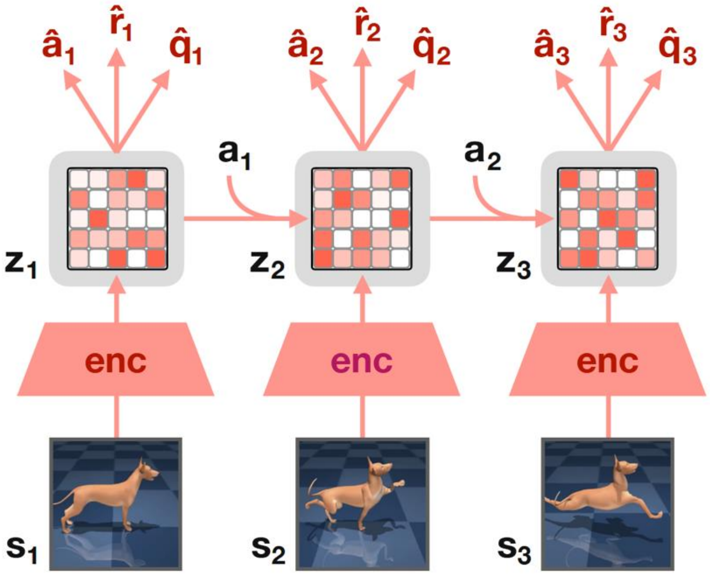
Hafner et al. Mastering diverse domains through world models. Nature 2024

TD-MPC: Decoder-free World Model

- Finish everything in the latent space
  - Decoder-free architecture
  - Directly predict the actions

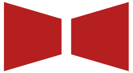






Hansen et al. Temporal difference learning for model predictive control. ICML 2022

Quick Summary

- Key Components in World Model
  - Autoencoder 
  - Sequential Model 
  - Reward Predictor (optional) 
- Three types of World Model
  - Dreamer
  - TD-MPC
  - MuZero

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3 World Model for Driving

Why We Need World Model

Driving requires decision-making at every moment

Current observations

Actions

Why We Need World Model

We need to know what will happen if we…

Generating Future given Driving Actions

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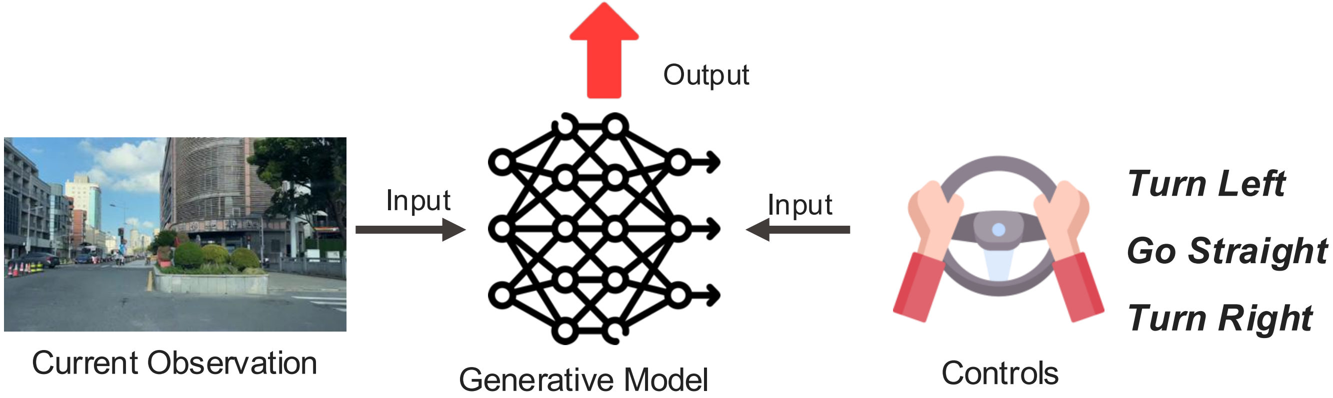
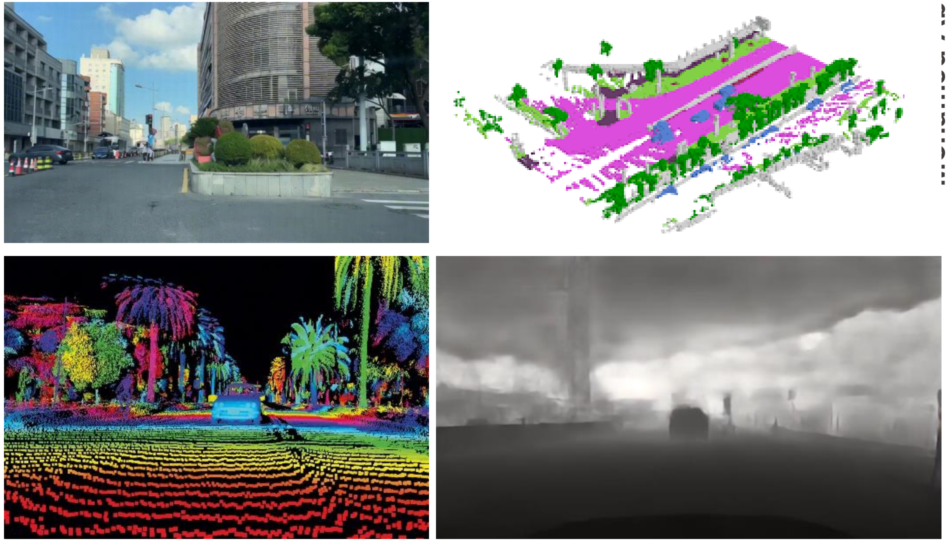


Gao et al. Vista: A generalizable driving world model with high fidelity and versatile controllability. NeurIPS 2024

World Model for Autonomous Driving

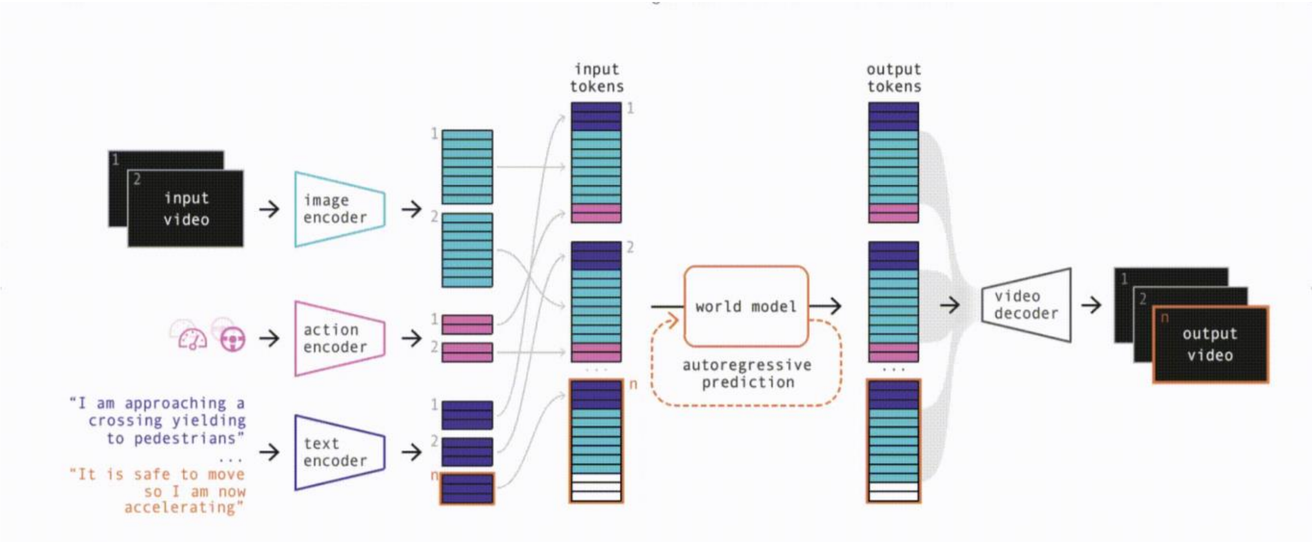
- What to generate
  - RGB Videos
  - Depth
  - Lidar
  - Occupancy / BEV
- How to generate
  - Diffusion Model
  - Autoregressive Transformer
  - Mixed



GAIA: The First Driving World Model based on Video Generation

Hu et al. Gaia-1: A generative world model for autonomous driving. arXiv preprint arXiv:2309.17080 (2023).

GAIA: Overview



Model architecture of GAIA-1

- Encode into discrete token
- Recurrently generate future token
- Decode token to videos

Hu et al. Gaia-1: A generative world model for autonomous driving. arXiv preprint arXiv:2309.17080 (2023).

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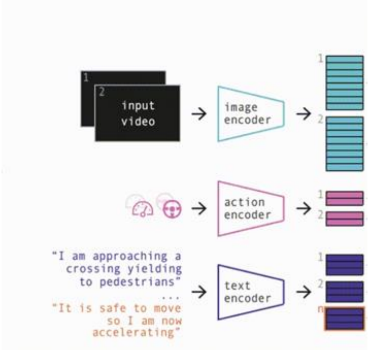
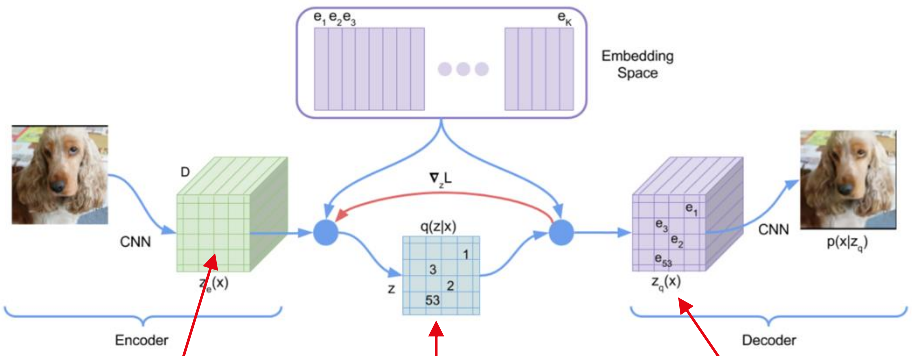
GAIA: Tokenization



Discrete!

A large dictionary



Encode into low-resolution feat

Look up a table and
replace the feature
with embedding

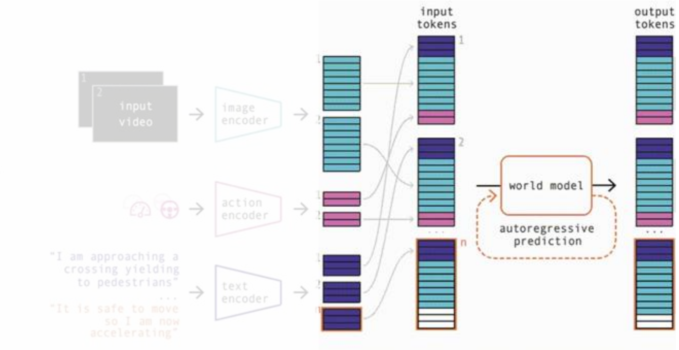
Decode the image

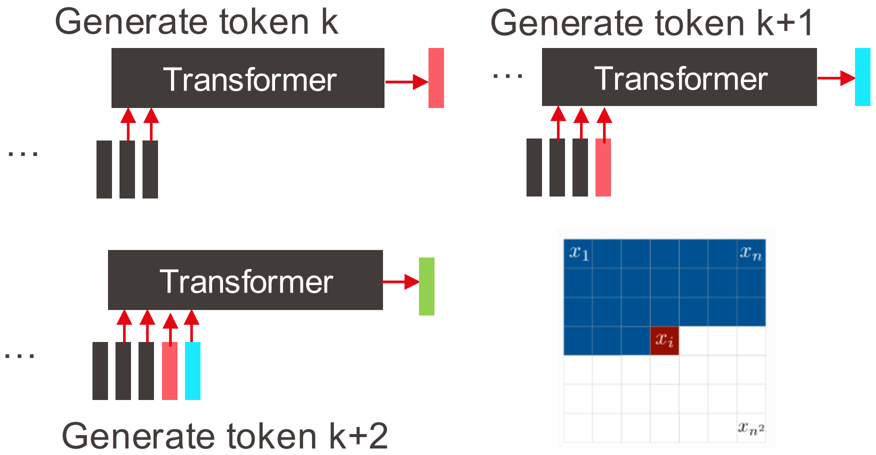
- Vector Quantized Variational Autoencoder (VQ-VAE)
  - Encode→Look Up & Replace →Decode

Hu et al. Gaia-1: A generative world model for autonomous driving. arXiv preprint arXiv:2309.17080 (2023).

Van et al. Neural discrete representation learning. NeurIPS 2017

GAIA: Autoregressive Prediction



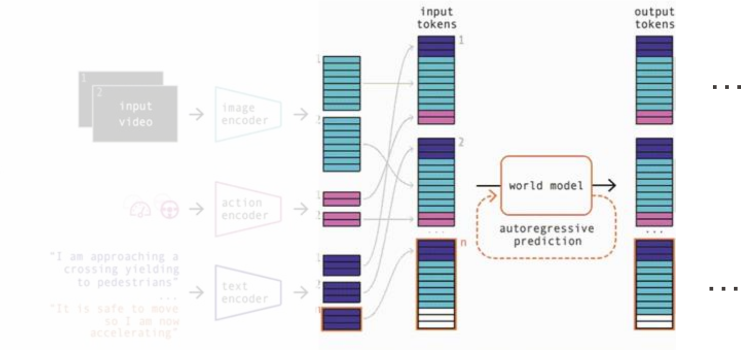


$$p(\mathbf{x}) = p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | x_1, \ldots, x_{i-1})$$

- GPT-like Autoregressive Prediction
  - Predicting the next token with previous tokens

Hu et al. Gaia-1: A generative world model for autonomous driving. arXiv preprint arXiv:2309.17080 (2023).

GAIA: Autoregressive Prediction

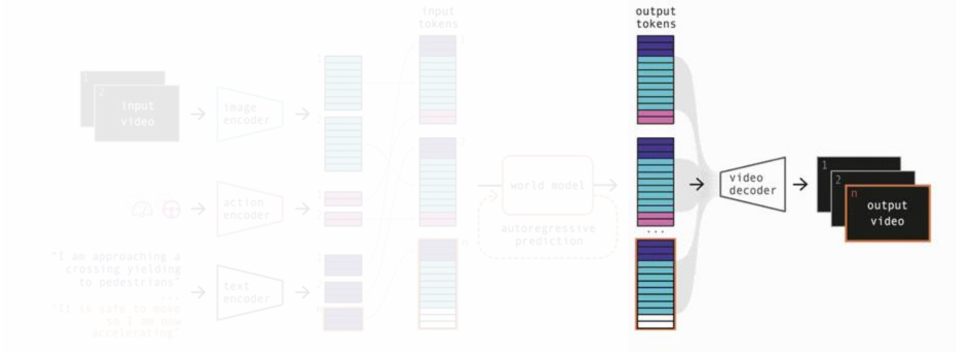


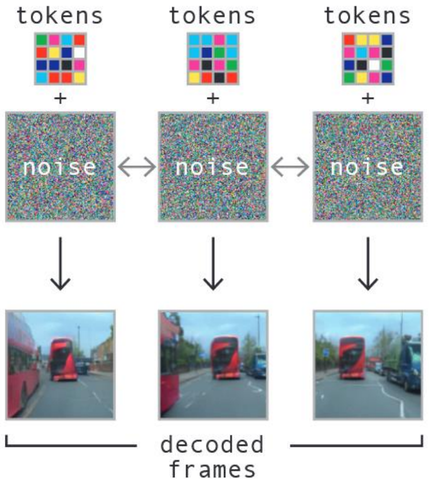
$$p(\mathbf{x}) = p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | x_1, \ldots, x_{i-1})$$

- GPT-like Autoregressive Prediction
  - Predicting t-th token with previous tokens

Hu et al. Gaia-1: A generative world model for autonomous driving. arXiv preprint arXiv:2309.17080 (2023).

GAIA: The First Driving World Model





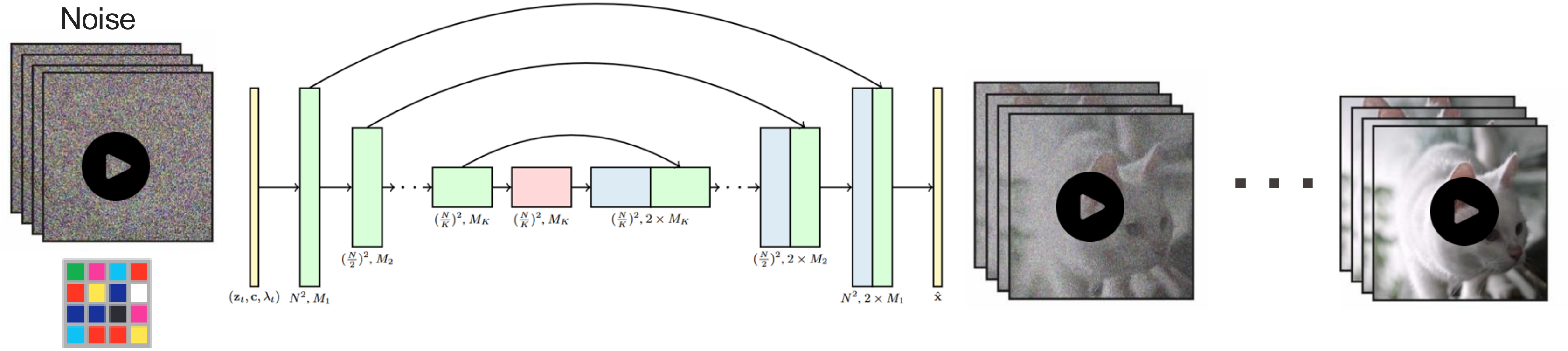
- Video Diffusion Decoder
  - Generate videos with diffusion

Hu et al. "Gaia-1: A generative world model for autonomous driving." arXiv preprint arXiv:2309.17080 (2023).

Preliminaries: Video Diffusion Model

- Generate videos with gradual denoising
  - Architecture: 3D U-Net

Noise



Condition ← Tell the model what it should generate

Ho et al. Video diffusion models. NeurIPS 2022

GAIA: The First Driving World Model

- Context & Short Text Control

Hu et al. "Gaia-1: A generative world model for autonomous driving." arXiv preprint arXiv:2309.17080 (2023).

GAIA: The First Driving World Model

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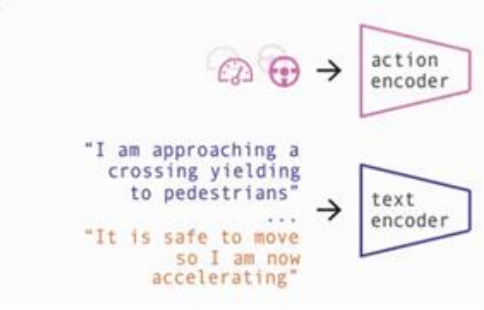
- Limitation in GAIA
  - Low-resolution (288 × 512)
  - Limited controls (text)



Vista: Higher Resolution & More Controls

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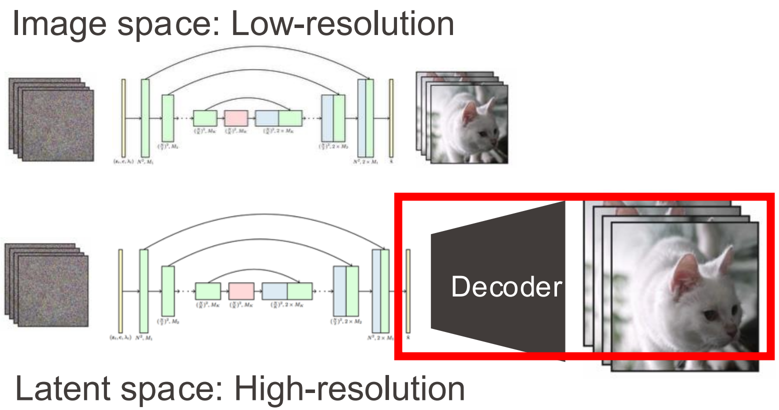
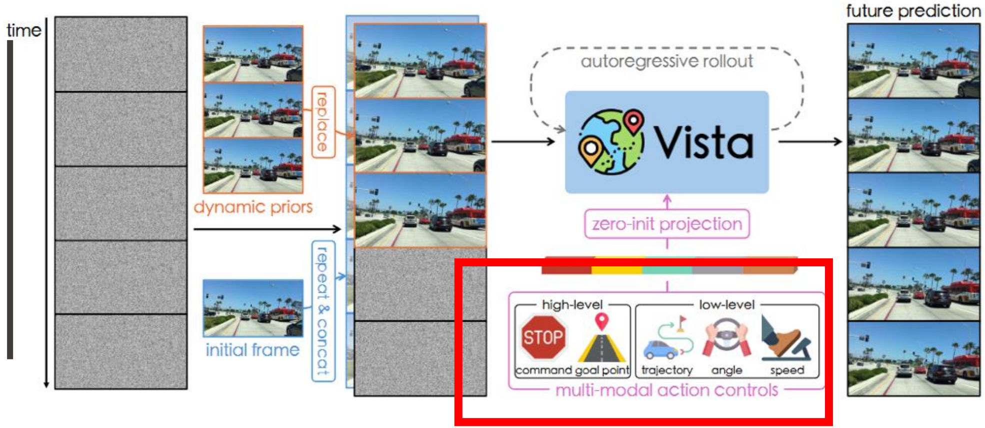
Vista: Two Technical Improvements





- Left: Latent video diffusion model
- Right: Inject visual controls (e.g., trajectory, target point etc.)

Rombach et al. High-resolution image synthesis with latent diffusion models. *CVPR 2022*

Vista: Higher Resolution & More Control

- Limitation in Vista
  - Low-dimensional generation lacking 3D understanding
  - Limited controls of environment

2D video

Large Gap

3D scene

Rombach et al. High-resolution image synthesis with latent diffusion models. *CVPR 2022*

GEM: Multi-modal Future Generation

Hassan et al. GEM: A Generalizable Ego-Vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control. CVPR 2025

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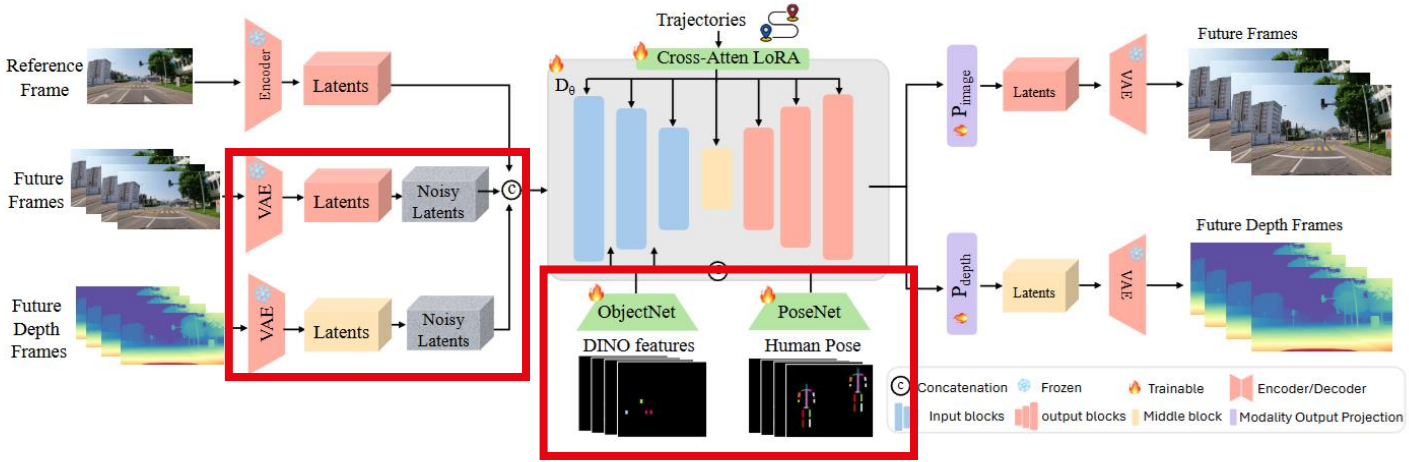
GEM: Two Technical Improvements

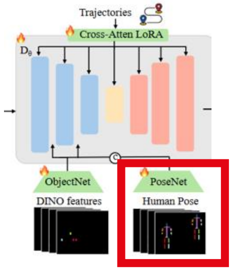


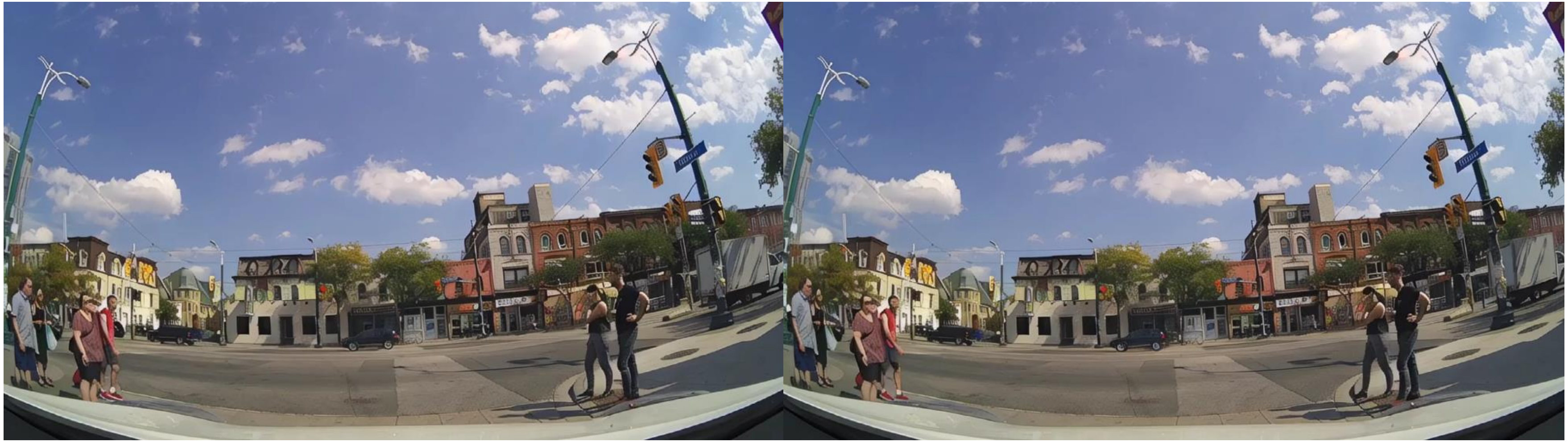
- Generate videos and corresponding depth
- Control the environment with diverse visual guidance

Hassan et al. GEM: A Generalizable Ego-Vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control. CVPR 2025

GEM: Environment Controls

- Human behaviors
  - E.g., pedestrians cross the road

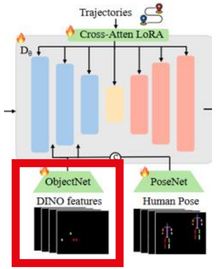




Hassan et al. GEM: A Generalizable Ego-Vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control. CVPR 2025

GEM: Environment Controls

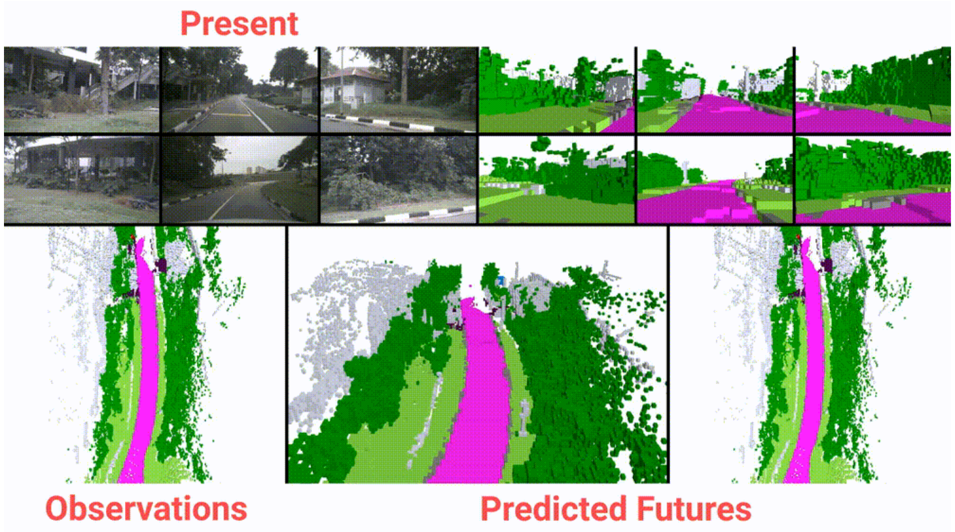
- Actions of other objects
  - E.g., a car appearing in the blue area

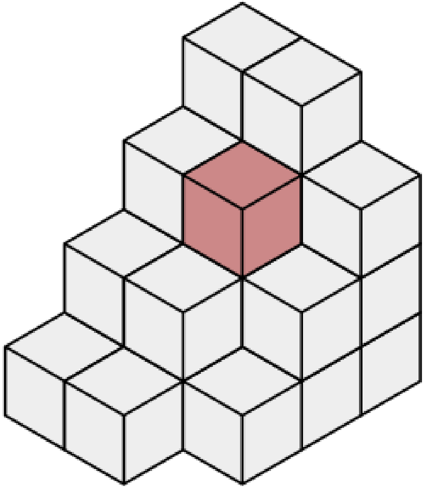


Hassan et al. GEM: A Generalizable Ego-Vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control. CVPR 2025

More Interesting World Models

- Generate 3D occupancy (Voxels)





Zheng et al. Occworld: Learning a 3d occupancy world model for autonomous driving. ECCV 2024.

More Interesting World Models

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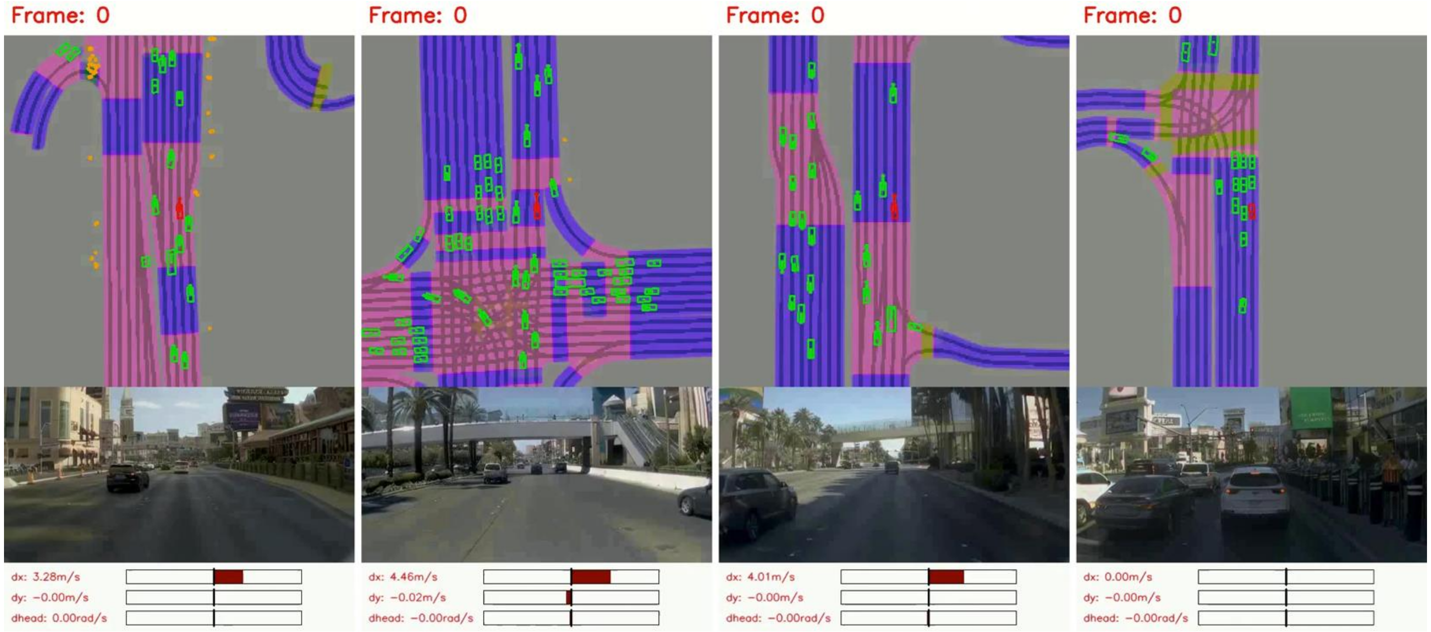
- Generate bird's-eye-view maps



Wu et al. Generating Multimodal Driving Scenes via Next-Scene Prediction. CVPR 2025

More Interesting World Models

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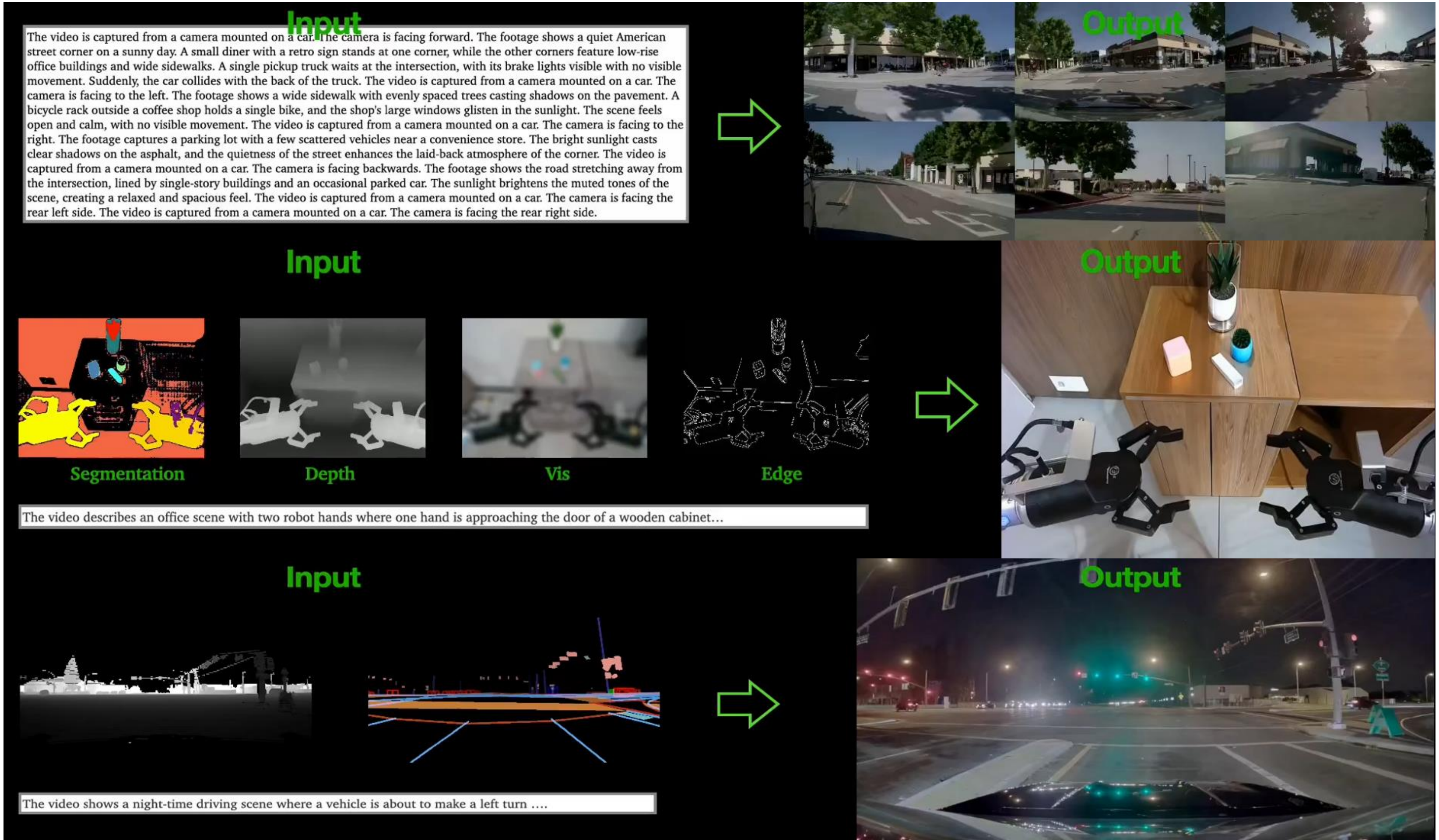
- Control the generation with vectorized maps

Wang et al. DriveDreamer: Towards Real-World-Drive World Models for Autonomous Driving. ECCV 2024

COSMOS: Cutting-Edge World Model

COSMOS: Cutting-Edge World Model



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World Model for Driving

1 Delving into World Model

2 Classic World Model

3 World Model for Driving

4 Takeaway Message

Conclusion

- World Model
  - Predict the future given the current observation and action
- Basic Components
  - Autoencoder
  - Sequential Model
  - Predictor
- Basic Generative Pipeline
  - Diffusion Model
  - Autoregressive Model
  - Mixed

Conclusion

https://www.youtube.com/watch?v=tx3DgeqHO5s

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Thank you for your attention!

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